Mathematics practice booklet: Grade 11 expectations

Grade 11 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Mathematics at Grade 11 level, following the Common Core State Standards for Mathematics standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Patterns, equations and algebra thinking

Standards practised: A-SSE.B.3, A-APR.B.2, A-APR.B.3, A-APR.D.6, A-REI.A.2, F-BF.A.1, F-BF.B.3, F-BF.B.4, F-LE.A.2, F-LE.A.4

A worked example

Solve 3^x = 20. Give the answer to 2 decimal places.

1. Take the logarithm of both sides: log(3^x) = log(20).
2. Bring the power down: x log(3) = log(20).
3. Divide: x = log(20) / log(3).
4. Use a calculator: x = 1.3010 / 0.4771, which is about 2.73.
5. Check: 3^2.73 is about 20.1, close to 20.

x is about 2.73

Questions

1. Factor: x^2 - 5x - 14

   A. (x + 7)(x - 2)

   B. (x - 7)(x + 2)

   C. (x - 14)(x + 1)

   D. (x - 7)(x - 2)

2. The graph of y = x^2 is moved to give g(x) = (x - 3)^2 + 4. Which move is this?

   A. 3 units left and 4 units up

   B. 3 units right and 4 units down

   C. 3 units right and 4 units up

   D. 4 units right and 3 units up

3. What is the remainder when P(x) = x^3 - 2x^2 + x - 6 is divided by x - 2?

   A. 4

   B. -6

   C. -24

   D. -4

4. Show that x = 1 is a zero of x^3 - 6x^2 + 11x - 6, then find the other zeros.

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5. Simplify, and say which values of x are not allowed: (x^2 - 9) / (x^2 + 3x)

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6. Solve: 2 / (x - 1) = 3 / (x + 2)

   A. x = 7

   B. x = 5

   C. x = 1

   D. x = -7

7. Solve and check: the square root of (x + 7) = x + 1

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8. A nursery in Ras Al Khaimah plants 400 mangrove seedlings and the number grows by 5% each year. Write a function for the number after t years and find the number after 10 years.

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9. Solve: log base 3 of x = 4

   A. x = 12

   B. x = 64

   C. x = 81

   D. x = 7

10. Rashid puts AED 2,000 in an account that grows by 10% each year. Use logarithms to find t when 2,000 × 1.1^t = 3,000, to 2 decimal places. In which year does the balance pass AED 3,000?

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11. Let f(x) = x + 2 and g(x) = x^2. What is f(g(3))?

    A. 11

    B. 25

    C. 9

    D. 5

12. A taxi fare in Dubai is C = 2k + 12 dirhams for k kilometres. Find the inverse formula that gives k from C. Then use it to find how far a passenger travelled for a fare of AED 40.

Section 2: Shapes, measuring and space

Standards practised: G-SRT.C.6, G-SRT.C.7, G-SRT.C.8, G-SRT.D.11, F-TF.A.1, F-TF.A.2, F-TF.B.5, F-TF.C.8

A worked example

From a point 100 m from the base of a tower in Abu Dhabi, the angle of elevation of the top is 40 degrees. Find the height of the tower to the nearest metre.

1. Draw a right triangle: the side next to the angle is 100 m and the side opposite is the height h.
2. Opposite and adjacent give the tangent: tan 40 degrees = h / 100.
3. h = 100 x tan 40 degrees = 100 × 0.8391.
4. h is about 83.9 m, which is 84 m to the nearest metre.

About 84 m

Questions

1. A right triangle has sides 5, 12 and 13 (13 is the longest). For the angle A opposite the side 5, what is tan A?

   A. 5 / 12

   B. 5 / 13

   C. 12 / 13

   D. 12 / 5

2. A 6 m ladder leans against a wall and makes an angle of 70 degrees with the ground. How high up the wall does it reach? Give the answer to 2 decimal places.

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3. A ramp rises 1.2 m over a horizontal distance of 10 m. Find the angle of the ramp with the ground, to 1 decimal place.

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4. Which is 150 degrees in radians?

   A. 2 pi / 3

   B. 5 pi / 6

   C. 3 pi / 4

   D. 5 pi / 3

5. A Ferris wheel has radius 20 m. How far does a seat travel along the circle as the wheel turns through pi / 3 radians? Give the answer to 1 decimal place.

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6. What is the exact value of tan 60 degrees?

   A. 1 / 2

   B. the square root of 3 / 2

   C. the square root of 3

   D. 1 / the square root of 3

7. On the unit circle, give the coordinates of the points at angles 90 degrees, 180 degrees and 120 degrees.

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8. What is the period of y = 2 sin(3x)?

   A. 2

   B. 2 pi

   C. 6 pi

   D. 2 pi / 3

9. The angle theta is in the second quadrant and sin theta = 3/5. What is cos theta?

   A. 4/5

   B. -4/5

   C. -3/4

   D. 3/4

10. In triangle ABC, angle A = 40 degrees, angle B = 60 degrees and side a = 8 cm (opposite A). What is side b, using the law of sines?

    A. 5.9 cm

    B. 12.4 cm

    C. 10.8 cm

    D. 8.0 cm

11. Two sides of a triangular garden are 7 m and 9 m, and the angle between them is 50 degrees. Use the law of cosines to find the third side, to 1 decimal place.

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12. Salem says that sin 2x is always twice sin x. Test x = 30 degrees and decide whether he is right.

Section 3: Data, graphs and chance

Standards practised: S-ID.A.2, S-IC.A.1, S-IC.B.3, S-IC.B.4, S-IC.B.5, S-ID.A.4

A worked example

The marks on a school test are roughly normal with mean 72 and standard deviation 8. About what percentage of students scored between 64 and 88?

1. 64 is one standard deviation below the mean (72 - 8) and 88 is two above (72 + 16).
2. In a normal distribution about 34% lies from the mean to 1 standard deviation, and about 47.5% from the mean to 2 standard deviations.
3. Add the two parts: 34% + 47.5% = 81.5%.

About 81.5%

Questions

1. Find the mean and the standard deviation of the data 4, 6, 8, 10, 12. Give the standard deviation to 2 decimal places.

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2. A school in Sharjah has 1,200 students. The principal asks 50 of them about the canteen. What is the population?

   A. The 50 students asked

   B. All 1,200 students

   C. The canteen menu

   D. The principal

3. Hamdan wants to know how many hours the students in his school sleep. He asks only the students who play in the school football team. Is this a good sample? Suggest a better way.

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4. In a random sample of 120 students, 36 walk to school. About how many of the 1,500 students in the school walk?

   A. 30

   B. 36

   C. 450

   D. 1,080

5. A poll of a random sample finds that 62% prefer a new park design, with a margin of error of 4%. Between which percentages is the true value likely to lie?

   A. 62% and 66%

   B. 56% and 68%

   C. 58% and 62%

   D. 58% and 66%

6. Poll A asks 100 people and poll B asks 1,000 people, both chosen at random. Which usually has the smaller margin of error, and why?

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7. A study gives 30 plants a new fertiliser and 30 similar plants none. The plants are assigned to the two groups at random. Is this an experiment or an observational study? Why is the random assignment important?

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8. After 4 weeks the fertilised plants have a mean height of 14.2 cm and the others 11.8 cm. What is the difference? What else would you want to know before deciding the fertiliser works?

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9. Test scores are roughly normal with mean 70 and standard deviation 10. About what percentage of scores lie between 60 and 80?

   A. 68%

   B. 34%

   C. 50%

   D. 95%

10. For the same scores (mean 70, standard deviation 10), about what percentage are above 90?

    A. 5%

    B. 2.5%

    C. 16%

    D. 34%

11. Using the same distribution (mean 70, standard deviation 10), find the z-score of 85. Layla has a mark with z-score -2: what is her mark?

Section 4: Numbers and calculating

Standards practised: N-CN.A.1, N-CN.A.2, N-CN.C.7, N-RN.A.1, N-RN.A.2, F-BF.A.2, A-SSE.B.4

A worked example

Write (2 + 3i)(1 - i) in the form a + bi.

1. Multiply each part: 2 × 1 + 2 x (-i) + 3i x 1 + 3i x (-i).
2. That is 2 - 2i + 3i - 3i^2.
3. Since i^2 = -1, the term -3i^2 equals +3.
4. Collect: (2 + 3) + (-2i + 3i) = 5 + i.

5 + i

Questions

1. Simplify the square root of -49 using i.

   A. -7

   B. 7

   C. 49i

   D. 7i

2. Add: (3 + 2i) + (5 - 4i)

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3. Multiply: (4 + i)(2 - 3i)

   A. 11 - 10i

   B. 5 - 10i

   C. 8 - 10i

   D. 6 - 2i

4. What is i^7 in simplest form?

   A. i

   B. -i

   C. 1

   D. -1

5. Write the complex conjugate of 4 - 3i and find the product of the number and its conjugate.

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6. Divide and write in the form a + bi: (6 + 2i) / (1 + i)

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7. Solve x^2 + 2x + 5 = 0.

   A. x = -1 ± 4i

   B. x = 1 ± 2i

   C. x = -1 ± 2i

   D. x = -2 ± 4i

8. What is 16^(3/4)?

   A. 12

   B. 8

   C. 64

   D. 4

9. Write the cube root of x^5 using a rational exponent, and simplify x^(1/2) times x^(3/2).

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10. A hall in Al Ain has 5 seats in row 1, and every row has 4 more seats than the row before. Write a rule for the number of seats in row n and find the number in row 20.

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11. A video is shared by 3 people on day 1, and the number of new shares doubles every day (3, 6, 12, 24). Find the common ratio and the number of new shares on day 8.

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12. Maryam saves AED 5 on day 1 and doubles the amount she saves each day for 7 days (5, 10, 20, and so on). How much has she saved in total? Why does this grow faster than saving AED 5 more each day?

Answer guide for parents

Section 1: Patterns, equations and algebra thinking

1. B. (x - 7)(x + 2) -7 × 2 = -14 and -7 + 2 = -5, so the factors are (x - 7)(x + 2). If your child chose A (x + 7)(x - 2), they swapped the signs, which gives +5x instead of -5x.
2. C. 3 units right and 4 units up. (x - 3) moves the graph 3 units right and + 4 moves it 4 units up. If your child chose A (3 left, 4 up), they read x - 3 as a move to the left.
3. D. -4. The remainder is P(2) = 8 - 8 + 2 - 6 = -4. If your child chose A (4), they lost the minus sign at the end.
4. Test x = 1: 1 - 6 + 11 - 6 = 0, so x - 1 is a factor. Dividing gives x^2 - 5x + 6 = (x - 2)(x - 3), so the zeros are 1, 2 and 3.
5. Factor to (x - 3)(x + 3) / (x(x + 3)) and cancel x + 3 to get (x - 3) / x. The bottom cannot be zero, so x cannot be 0 or -3.
6. A. x = 7. Cross multiply: 2(x + 2) = 3(x - 1), so 2x + 4 = 3x - 3 and x = 7. If your child chose D (x = -7), they moved terms across with the wrong signs.
7. Square both sides: x + 7 = x^2 + 2x + 1, so x^2 + x - 6 = 0 and x = 2 or x = -3. Check each in the original: x = 2 gives 3 = 3, but x = -3 gives 2 = -2, so only x = 2 works.
8. Each year multiplies by 1.05, so N = 400 × 1.05^t. After 10 years, N = 400 × 1.6289, which is about 652 seedlings.
9. C. x = 81. log base 3 of x = 4 means 3^4 = x, so x = 81. If your child chose A (12), they multiplied 3 × 4 instead of raising 3 to the power 4.
10. Divide by 2,000: 1.1^t = 1.5, so t = log(1.5) / log(1.1), about 4.25. After 4 years the balance is about AED 2,928 and after 5 years about AED 3,221, so it passes AED 3,000 in the 5th year.
11. A. 11. Work from the inside: g(3) = 9, then f(9) = 9 + 2 = 11. If your child chose B (25), they did f first and then g, which finds g(f(3)).
12. To undo C = 2k + 12, subtract 12 and then divide by 2: k = (C - 12) / 2. For C = 40, k = 28 / 2 = 14 km.

Section 2: Shapes, measuring and space

1. A. 5 / 12. tan A = opposite / adjacent = 5 / 12. If your child chose B (5 / 13), they used the longest side, which gives sin A.
2. The wall height is opposite the 70 degree angle and the ladder is the hypotenuse, so use sine: 6 x sin 70 degrees, which is about 5.64 m.
3. The rise 1.2 is opposite the angle and 10 is next to it, so tan of the angle = 1.2 / 10 = 0.12. The angle is about 6.8 degrees.
4. B. 5 pi / 6. 150 x pi / 180 = 5 pi / 6, because 180 degrees is pi radians. If your child chose D (5 pi / 3), they divided by 90 instead of 180.
5. Arc length = radius x angle in radians = 20 x pi / 3, which is about 20.9 m.
6. C. the square root of 3. tan 60 = sin 60 / cos 60 = (the square root of 3 / 2) / (1 / 2) = the square root of 3. If your child chose B (the square root of 3 / 2), they gave sin 60 and forgot to divide by cos 60.
7. The point at angle t is (cos t, sin t). So 90 degrees is (0, 1), 180 degrees is (-1, 0), and 120 degrees is (-1/2, the square root of 3 divided by 2).

8. D. 2 pi / 3. The period of sin(bx) is 2 pi divided by b, so 2 pi / 3. If your child chose C (6 pi), they multiplied 2 pi by 3 instead of dividing.
9. B. -4/5. sin^2 + cos^2 = 1 gives cos^2 = 16/25, and cosine is negative in the second quadrant, so -4/5. If your child chose A (4/5), they forgot the sign.
10. C. 10.8 cm. The law of sines gives b = 8 x sin 60 / sin 40 = 8 × 0.866 / 0.643, about 10.8 cm. If your child chose A (5.9 cm), they flipped the fraction to sin 40 / sin 60.
11. The law of cosines gives c^2 = 7^2 + 9^2 - 2 × 7 × 9 x cos 50 = 130 - 126 × 0.643, about 49.0. So c is about 7.0 m.
12. No. sin 60 degrees is about 0.866, but 2 x sin 30 degrees = 2 × 0.5 = 1. One example that does not match is enough to show the rule is false.

Section 3: Data, graphs and chance

1. Mean = (4 + 6 + 8 + 10 + 12) / 5 = 8. Squared distances 16, 4, 0, 4, 16 add to 40; 40 / 5 = 8, and the square root of 8 is about 2.83.
2. B. All 1,200 students. The population is everyone we want to know about: all 1,200 students. If your child chose A (the 50 students asked), they named the sample instead.
3. No, it is biased, because athletes may not sleep like everyone else. Better: pick students at random, for example by drawing random numbers from the class register.
4. C. 450. 36 / 120 = 30%, and 30% of 1,500 is 450 students. If your child chose B (36), they used the sample count and did not scale up to the whole school.
5. D. 58% and 66%. 62% minus 4% is 58% and 62% plus 4% is 66%. If your child chose A (62% and 66%), they added the margin but forgot to subtract it.
6. Poll B. A larger random sample varies less from one sample to another, so its estimate is more precise and its margin of error is smaller.
7. It is an experiment, because the researcher gives the fertiliser. Random assignment makes the two groups alike on average, so a difference in growth is more likely caused by the fertiliser.
8. 14.2 - 11.8 = 2.4 cm. You would also want the spread of heights within each group, and to check whether 2.4 cm is bigger than chance variation, for example by repeating the experiment.
9. A. 68%. 60 and 80 are 1 standard deviation either side of 70, and that range holds about 68% of the scores. If your child chose D (95%), they used 2 standard deviations.
10. B. 2.5%. 90 is 2 standard deviations above the mean. 95% lie within 2, so 5% is split between two tails, leaving 2.5% above. If your child chose A (5%), they forgot to halve it.
11. z = (85 - 70) / 10 = 1.5, so 85 is 1.5 standard deviations above the mean. A z-score of -2 is 2 standard deviations below: 70 - 2 × 10 = 50.

Section 4: Numbers and calculating

1. D. 7i. The square root of -49 is the square root of 49 times the square root of -1, which is 7i. If your child chose A (-7), they ignored i and moved the minus sign onto the 7.
2. Add the real parts: 3 + 5 = 8. Add the imaginary parts: 2i - 4i = -2i. The answer is 8 - 2i.
3. A. 11 - 10i. 8 - 12i + 2i - 3i^2 = 8 - 10i + 3 = 11 - 10i, because i^2 = -1. If your child chose C (8 - 10i), they forgot that -3i^2 becomes +3.
4. B. -i. i^4 = 1, so i^7 = i^4 x i^3 = i^3 = -i. If your child chose A (i), they took i^3 to be i instead of -i.

5. The conjugate changes the sign of the imaginary part: 4 + 3i. The product is (4 - 3i)(4 + 3i) = 16 + 9 = 25, a real number.
6. Multiply top and bottom by 1 - i. The top is (6 + 2i)(1 - i) = 8 - 4i and the bottom is 1 + 1 = 2, so the answer is 4 - 2i.
7. C. x = -1 ± 2i. The discriminant is 4 - 20 = -16 and its square root is 4i, so x = (-2 ± 4i) / 2 = -1 ± 2i. If your child chose D (-2 ± 4i), they forgot to divide everything by 2.
8. B. 8. The fourth root of 16 is 2, then 2^3 = 8. If your child chose A (12), they multiplied 16 by 3/4 instead of taking the root and the power.
9. The cube root is the power 1/3, so the cube root of x^5 is x^(5/3). Add the exponents: x^(1/2) times x^(3/2) = x^(1/2 + 3/2) = x^2.
10. Row 1 has 5 and each row adds 4, so a_n = 5 + 4(n - 1). Row 20 has 5 + 4 × 19 = 5 + 76 = 81 seats.
11. Each day is the day before times 2, so the ratio is 2. Day 8 has 3 × 2^7 = 3 × 128 = 384 new shares.
12. Total = 5 x (2^7 - 1) / (2 - 1) = 5 × 127 = AED 635. Doubling multiplies by the same number each day, so it overtakes adding AED 5 a day, which gives only 5 + 10 + … + 35 = AED 140.